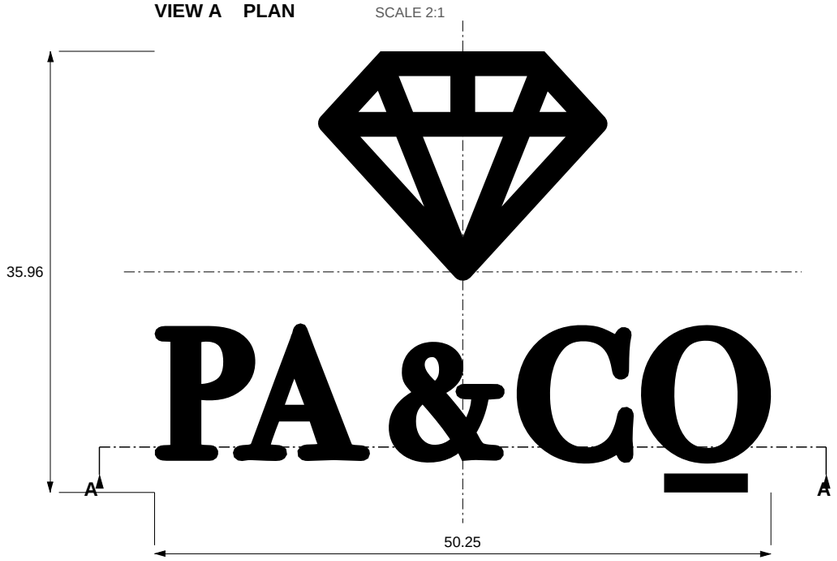
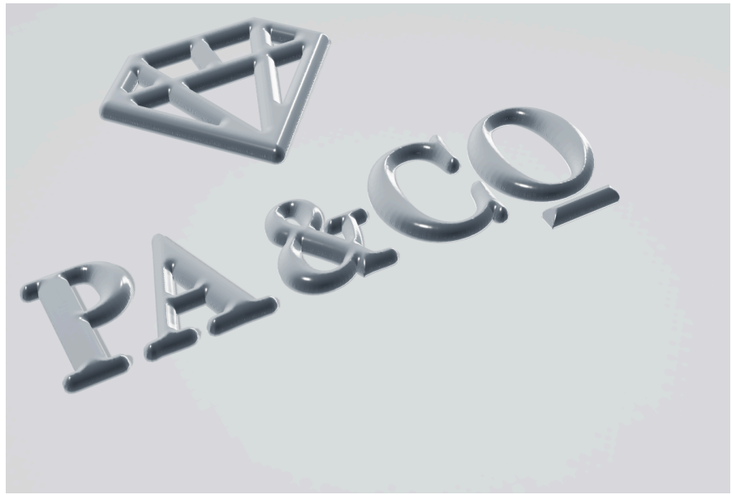
VIEW A PLAN

SCALE 2:1



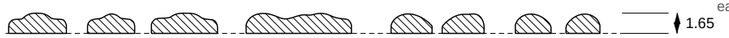


ISOMETRIC VIEW

SECTION A-A

SCALE 2:1

the strokes the plane passes through, cut at y = -14.28



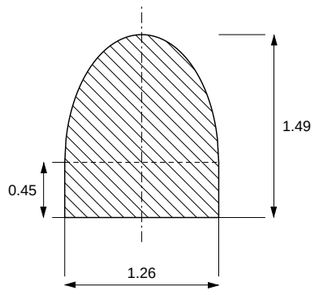
Crown varies with the width of the stroke the plane meets. See DETAIL C and the schedule on sheet 2.

DETAIL C STROKE PROFILES

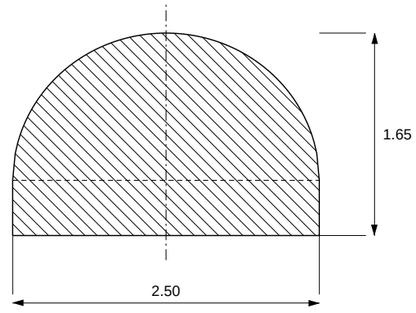
SCALE 20:1

the thinnest stroke in the artwork, and the width from which the dome is full height

each stroke is domed to its own width, so both close horizontal at the centre



THINNEST 1.26

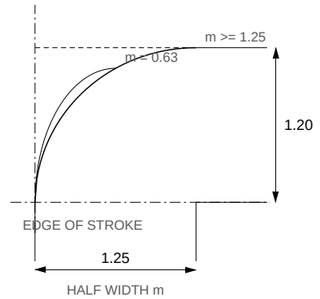


FULL HEIGHT 2.50

The dome is fitted to the width of the stroke it is on, not struck on one radius for the whole part. A narrow stroke reaches a lower crown, but it still arrives at its own centre horizontal, so the top surface is continuous everywhere. There is no ridge, and no edge anywhere on the part except the outline itself.

DETAIL B DOME PROFILE

fitted to the width of the stroke it is on



$z = C \times \sqrt{2t - t^2}$, $t = d / m$

C = 1.20 for m >= 1.25, otherwise C = 1.20 x sqrt(2u - u^2), u = m / 1.25.
d is measured in plan from the edge of the stroke and m is half the width of that stroke, so every stroke reaches its centre with the surface horizontal. One radius for the whole part put a ridge down the middle of every narrow stroke; this does not.

CROWN HEIGHT BY STROKE WIDTH

STROKE WIDTH	CROWN	NOTE
1.26	1.49	thinnest in artwork
1.50	1.55	
2.00	1.63	
2.50	1.65	full height from here
3.42	1.65	widest point in artwork

Crown figures include the 0.45 mm base.

NOTES

1. All dimensions in millimetres. Do not scale from this drawing.
2. 7 separate bodies. They do not touch and must not be joined. Positions and gaps on sheet 2.
3. Outline is fixed. No stroke may be thinned; 1.26 mm is the minimum and is already at the limit.
4. Tightest gap between bodies is 0.79 mm. See sheet 2.
5. Top surface domed per DETAIL B, fitted to each stroke width. No ridges. Underside flat, no draft.
6. Finish satin silver, uniform. No polished or mirror areas.
7. Geometry of record is the STL: paco-logo-SMOOTH-v3-dome1.20-base0.45.stl
8. General tolerance +/-0.05. Crown height +/-0.10.

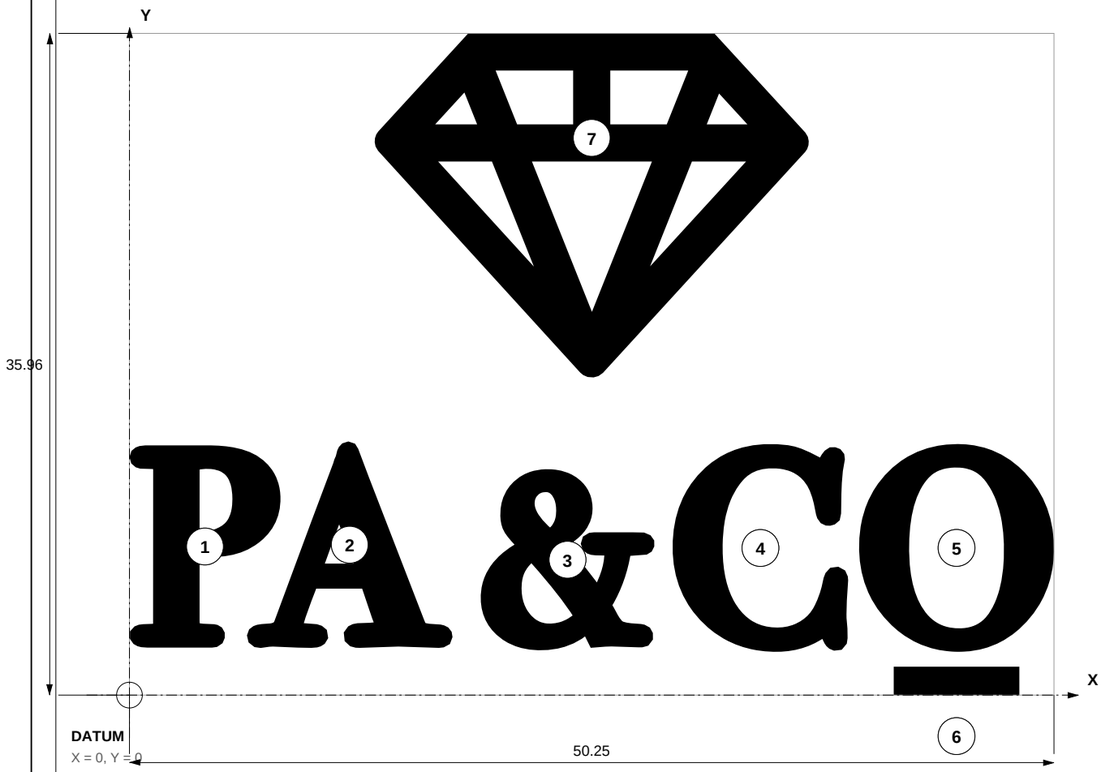
PA & CO LOGO, DOMED

THIRD ANGLE

MATERIAL	per supplier	UNITS	mm	DWG NO.	PACO-3D-02
FINISH	satin silver	MIN STROKE	1.26	SCALE	AS NOTED
OVERALL	50.25 x 35.96 x 1.65	BODIES	7	SHEET	1 OF 2
				REV	A

VIEW D PLAN, ITEM NUMBERS AND DATUM

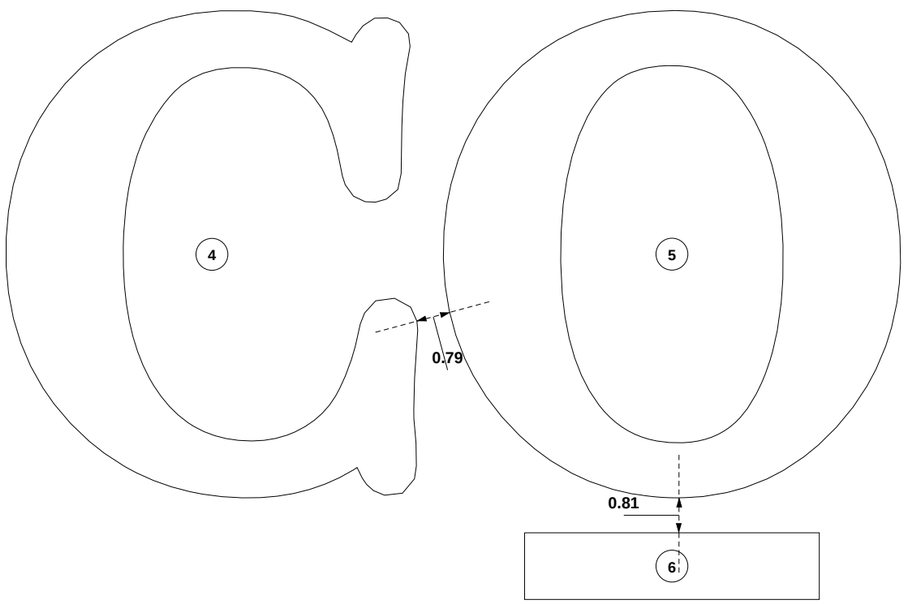
SCALE 3:1



DETAIL E THE TIGHTEST GAPS

SCALE 7.0:1

dimensioned between the two points that are actually closest, not between bounding boxes



ORDINATE SCHEDULE

X and Y to the lower-left corner of each body, from the datum

ITEM	BODY	X	Y	WIDTH	HEIGHT	CTR	CROWN
1	P	0.00	2.58	8.21	10.99	1	1.65
2	A	6.38	2.56	11.17	11.22	1	1.65
3	AMPERSAND	19.09	2.44	9.43	9.84	2	1.65
4	C	29.52	2.35	9.54	11.28	0	1.65
5	O	39.65	2.35	10.59	11.29	1	1.65
6	BAR UNDER O	41.54	0.00	6.83	1.54	0	1.56
7	DIAMOND	13.33	17.26	23.59	18.69	7	1.65

CTR is the number of enclosed counters in the body (the diamond's facets, the bowl of a P).
CROWN is the full height that body reaches, base included; a body narrower than 2.50 mm cannot reach 1.65.

CLEARANCE SCHEDULE

every pair of bodies within 4 mm, closest first

FROM		TO		GAP
4	C	5	O	**0.79**
5	O	6	BAR UNDER O	0.81
3	AMPERSAND	4	C	1.00
1	P	2	A	1.21
2	A	3	AMPERSAND	1.95
4	C	6	BAR UNDER O	2.84

These are gaps in the artwork, not assembly tolerances. The tightest one sets what the tooling has to hold.

STROKE SCHEDULE

measured on the medial axis of each body, terminals excluded
WIDEST is a pool where strokes meet, not a stroke; anything over 2.50 is at full height.

ITEM	BODY	THINNEST	WIDEST	AREA
1	P	1.27	2.70	49.2
2	A	1.32	2.92	49.3
3	AMPERSAND	**1.26**	2.38	53.7
4	C	1.30	2.73	49.5
5	O	1.27	2.73	57.8
6	BAR UNDER O	1.57	1.57	10.5
7	DIAMOND	1.63	3.42	201.4

AREA is the plan area of the body in mm2, counters deducted. Total 471.6 mm2.

PA & CO LOGO, DOMED

THIRD ANGLE

				DWG NO.	PACO-3D-02
MATERIAL	per supplier	UNITS	mm	SCALE	AS NOTED
FINISH	satin silver	MIN STROKE	1.26	SHEET	2 OF 2
OVERALL	50.25 x 35.96 x 1.65	BODIES	7	REV	A